

CS471 – Lesson 12 In-Class Worksheet

Markov Decision Processes II: Values and Policies

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Define policies, interpret expected cumulative rewards, and evaluate value functions.

1. Policies

An autonomous Space Force satellite is tasked with monitoring a designated sector.

- a. Circle the best definition of a policy (π):
 - A. A rule that maps each state to an action or action distribution.
 - B. A list of all rewards received during one episode.
 - C. A model that gives the probability of every next state.
 - D. A function that stores only immediate rewards.
- b. Define a **deterministic policy** versus a **stochastic policy**.

2. State Value and Action Value

Complete each mathematical statement below using $V^\pi(s)$ or $Q^\pi(s, a)$.

- Expected return starting in state s and then following policy π : _____
- Expected return after taking action a in state s and then following policy π : _____

Question: Which of these two quantities compares different actions available in the same state?

Question: What does an **optimal state-value function** ($V^*(s)$) represent?

3. Discounted Return (G)

The Space Force satellite receives an immediate reward of +2 now for re-positioning, and a reward of +6 one step later for successfully taking an image. Let the discount factor $\gamma = 0.5$.

Question: Write the two-step expected return equation and compute its value (G). What mathematical and operational effect does setting $\gamma < 1$ have on later rewards?

Part 2: Student Independent Practice

Objective: Apply discounted return equations and evaluate greedy action-value selections in tactical scenarios.

A. Interpreting Action-Values (Q-Values)

A cyber-defense agent is actively monitoring a network node under high load. The agent calculates the following action values for its current state (s) based on its current security policy (π):

- $Q^\pi(s, \text{Block}) = 8$
- $Q^\pi(s, \text{Monitor}) = 5$
- $Q^\pi(s, \text{Ignore}) = 2$

1. If the agent utilizes a **greedy choice**, which action will it select?
2. What exactly does the largest Q-value (8) represent in the context of this MDP?

B. Comparing Returns

An Autonomous Ground Vehicle (AGV) must clear a path through a contested urban environment. It has two options:

- **Action A (Direct Route):** Gives an immediate reward of +4 and successfully ends the route.
- **Action B (Covert Route):** Gives a reward of +1 now, and a reward of +4 on the next time step.

Given a discount factor of $\gamma = 0.5$:

1. Compute the expected return for Action A.
2. Compute the expected return for Action B.

- 3.** Compare the returns and identify the preferred action for the AGV to take.